POLITECNICO MILANO 1863
DEPARTMENT OF AEROSPACE SCIENCE AND TECHNOLOGY

FLY MI

13TH JUNE 2025

FLIGHT READINESS REVIEW

13 - Fly-Mi - Politecnico di Milano

TEAM MEMBERS

Marta Catalano, *BSc Aerospace Engineering, 3rd year*
Filippo Coacci, *MSc Aeronautical Engineering, 2nd year*
Luigi Hayashida, *BSc Aerospace Engineering, 3rd year*
Paola Mansueti, *BSc Aerospace Engineering, 3rd year*
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Leonardo Salvini, MSc High Performance Computing, *1st year*
Lorenzo Martella, *BSc Mechanical Engineering, 3rd year*
Domenico Lo Faro, *BSc Aerospace Engineering, 3rd year*
Giuseppe Rizi, *MSc Aeronautical Engineering, 1st year*
Luca Baldi, *BSc Aerospace Engineering, 2nd year*

ACADEMIC LEAD
Lorenzo Trainelli

TEAM LEADER

FRR REVIEWER AND UAS PILOT

Flight Readiness Review

Fly-Mi

Header

Team: Fly-Mi

Team Number: 13

University: Politecnico di Milano

Type of Meeting: Flight Readiness Review

Location: Online, Microsoft Teams

Participants

Present:

- Giulio Ricotti, Chartered Engineer & UAS Pilot
- Paola Mansueti, team member
- Lorenzo Martella, team member
- Luigi Hayashida, team member
- Federico Sottocorno, Fly-Mi associate
- Federico Zacco, Fly-Mi associate

Agenda

1. Significant changes from Design & Development specification review
2. Corrective actions required from Design & Development specification feedback review
3. Critical systems testing review
4. Pre-flight checklist presentation

Introduction

As the manufacturing and system testing ended, the Flight Readiness Review is conducted as specified in Annex D.5 to verify the airworthiness of the aircraft and that is ready to undertake flight testing. This review is being conducted by the chartered engineer and UAS pilot Giulio Ricotti, who supervised the control system flight testing campaign and worked together with the team as advisor and mentor. The agenda follows the specifications outlined in Annex D.5 and serves as a complementary document to the IMechE Aero Form 701 and the Declaration of Conformity, providing detailed descriptions of the points requiring further elaboration.

The following document presents the minutes of this meeting, organized in table format for greater clarity.

Discussion

- **Topic 1:** Significant changes from Design & Development specification review
  - Summary:

Significant Change from DDS	Possible Impact on Safety and Airworthiness	Reviewer notes
DDS 4.3.1 - Fuselage: Replacement of carbon fiber fuselage monocoque due to molds sponsor issues with interlocking internal structure in nomex honeycomb – carbon fabric sandwich bulkheads. Outer skin in balsa and oracover.	The interlocking structure adds more weight due to adhesives, causing a possible reduction in allowable payload weight. This solution though, is more accessible and modular, facilitating modifications and reinforcements that may be required during assembly or in competition. No impact on safety or Airworthiness of the aircraft	The most critical part of the new fuselage design is the central wingbox, because it bears the higher loads. The tube connecting the wings bends but, given its position, does not bear moment. The internal structure of the wingbox is made of Nomex reinforcement plates oriented to resist bending moment. During flight, the upper part of the wingbox is compressed and the lower part is under tension. To address this, the lower part was laminated using unidirectional (UD) carbon fibre and Rohacell was added to prevent compression of the upper part. The new fuselage design was validated using static and dynamic FEM simulations. The results show that there is no risk to safety or Airworthiness.
DDS 4.4.1 - Electronic Components Selection. It was stated that for the Ground Control Station, the Mateksys mR24-30 2.4GHz Receiver was not compliant with CE regualtion and needed replacement. This component won't be used, instead Mission Planner on Android tablet, connected to the radio transmitter through ExpressLRS Backpack + Yaapu telemetry on radio LCD screen will serve as ground station	No impact on Safety or Airworthiness	Using an Android tablet connected to the radio makes it easier to control the drone during flight. This gives better control of the drone, even when flying autonomously, th- ereby improving safety.
DDS 4.4.1 - Electronic Components Selection. Servo motor for the Payload Release Sytstem was not mentioned. A MKS HV6150 will be used for PRS	No impact on Safety or Airworthiness	/

– Assigned actions:

Action	Responsible	Deadline
No action was required after speaking with the reviewer.	/	/

- **Topic 2:** Corrective actions required from Design & Development specification feedback review

– Summary:

Corrective action required from DDS	Correction Description	Reviewer notes
Requirement Review: *You are required to have two separate safety links - one which removes power to the motors (4.2.3) and one which removes power to the whole aircraft (4.2.2). They aremovable links and not switches.*	The switches have been replaced with removable links.	The removable links are clearly visible and easily accessible.
Design: Main gear position vs CG position. *The main gear set-back position may require a significant nose up moment. We advise you to check that you have enough elevator control, or consider raking your landing gear forwards, if you cannot movethe rest of the structure*	The rear landing gear has been mounted in such a way as to prevent the aircraft from having a negative pitch during take-off. The engine power should be sufficient to ensure take-off.	The front and rear landing gear are mounted to prevent negative pitch. To ensure safety during landing, it is important to connect the landing gear as much as possible to the main structure. The rear landing gear should be screwed into the main structure. To prevent delamination or concentrated loads on the carbon fibre, a small aluminium plate should be located on the rear landing gear. This modification was implemented and checked.
Design: wing FEA and Structure. *In the image... ...of the FEA of your wing, ... you have two semi-spars which act to connect your wing segments together (...). We assume that the green element is your wing spar, and that it is split at the point shown by the full chord rib at about 50% span. The concern is that the structure shown has many discontinuities in it from a load path perspective. (...) You may wish to review your structural analysis and do some physical testing of your wing structure.*	In the FEM analyses, the two spars are separated and consistent with the real structure. The load path is realistically distributed. The wing structure was tested through a static test by lifting the aircraft from the wingtips, demonstrating the structural validity of the wing FEA Analysis.	/
Design: Flight Termination System. *You do not describe the Flight Termination System.*	The Flight termination system is exactly as prescribed by the regulation and follows the advanced failsafe settings on Ardupilot	

- Assigned actions:

- **Topic 3:** Critical Systems Testing Review
  - Summary:

System Tested	Description	Reviewer notes
Control System - System Familiarization	- Integrate and configure ArduPilot - Understand system-level interactions - Basic autopilot functions (e.g., loiter, GCS, trim calibration)	
Control System - Autopilot Tuning & Guidance	- Tune control loops on the actual model - Train and iterate mission procedures (modes transitions, checklists) - Enable basic waypoint navigation and vertical profile control	
Control System - Full Autonomous Operations	- Achieve autonomous takeoff and landing - Fine-tune ground handling, approach, and landing sequences - Final tuning for robust end-to-end autonomous flight	
Structures - Tensile test on the wing retention mechanisms.	First sample resisted up to 1000 Newton. The test of a second, smaller sample showed that the sample withstood more than the adhesive, which detached at 250 newtons. The components were therefore considered suitable for the loads.	Ensure the surface is porous to allow the adhesive to work properly.
Structures - Test of the payload release system	Before permanently closing the wingbox, the proper functioning of the servo and the strength of the pin under the load applied by the payload were evaluated through a ground test. After some adjustments to the servo positioning and its sizing, the mechanism proved to be valid and functional.	
Structures - Failure test on 3D-printed components	All 3D components went trought an interative process to optimise printing process parameters and weight.	
Structures - Failure test of the first landing gear built	The first rear landing gear failure test demonstrated the component's inability to withstand landing loads. A different layup sequence was therefore developed, which ensured the creation of much stronger parts.	The front landing gear should be as connected as possible to the main strutcure to ensure a good load distribution in landing

Structures - Test of the landing gear behaviour during taxiing	Once assembled, it was observed that the shock-absorbing nose wheel, when compressed under the weight of the aircraft, did not ensure a neutral attitude during taxiing. For this reason, along with the feedback received regarding the relative position between the landing gear and the center of gravity, the holes for the rear wheels were raised to ensure a neutral attitude in the case of maximum shock absorber compression, and thus a slightly nose-up attitude when not compressed, to facilitate rotation during takeoff.	
Structures - Static Test	Static test completed without issues, by lifting the aircraft from the wing tips, with a weight of 6.8 kg consisting of: - complete structure - no sensors - unfinished wiring - 2.5 kg extra weight	It's importante to ensure a good load distribution between the wing
Structures- Tail assembly	The two tail fins are connect to the main tail tube with square section carbon fibre tubes to ensure their position and preventing them from rotating. The tubes are screwed to the tail hub to lock axial movement. A load test on the tail was conducted and completed without issues.	
Structures- Tail control surfaces	The control surfaces were 3D-printed in two parts, which were then assembled using a carbon fibre tube that ran from the lower part to the upper part. The two parts were then glued together.The control surfaces are secured to the main structure at two different points on their end.	To prevent bending during abrupt manoeuvres, the control surfaces should also be secured to the main structure in the centre.

- Assigned actions:

Action	Responsible	Deadline
Ensure surfacety porosity	Assembly team	13th June 2025
Secure Tail control surfaces in the centre	Assembly team	13th June 2025

- **Topic 4:** Pre-flight checklist presentation
  - Summary:
    The following table resumes the pre-flight checklist presented by the team.

Pre flight checklist	Reviewer notes
Ardupilot parameter list and Mission waypoints	
Shake the wing to avoid pinching servo cables with the bayonet	
Check servo connector correctly inserted	
Tail connector fastened	

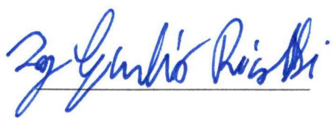
Condition check: ∗ Visual inspection of the plane (intact, clean and working components): - Landing gear (LG) - Structural integrity - Fuselage (no loose components) - Payload release correctly retained - Clean aerodynamic surfaces ∗ Securely tightened bolts, retention mechanisms and propeller ∗ Check connectors condition and electronic components correctly fastened ∗ Wing retention pin inserted ∗ Correct movement of all moving parts: - Servomotors - Control surfaces - Payload release system - Nose landing gear	
Batteries check (SOC and correct retention): - Radio Battery - Ground control station - Main Battery (main switch off)	
CG check through balances for correct stability margin	

Conclusion

At the close of the meeting, the reviewer confirmed compliance with the requirements and the airworthiness of the aircraft, stating that the drone is ready for takeoff. He expressed his approval of the work done so far and wished the team good luck and continued success, concluding the meeting.

Closure

Approved by:



(Reviewer)

On: 13/06/2026